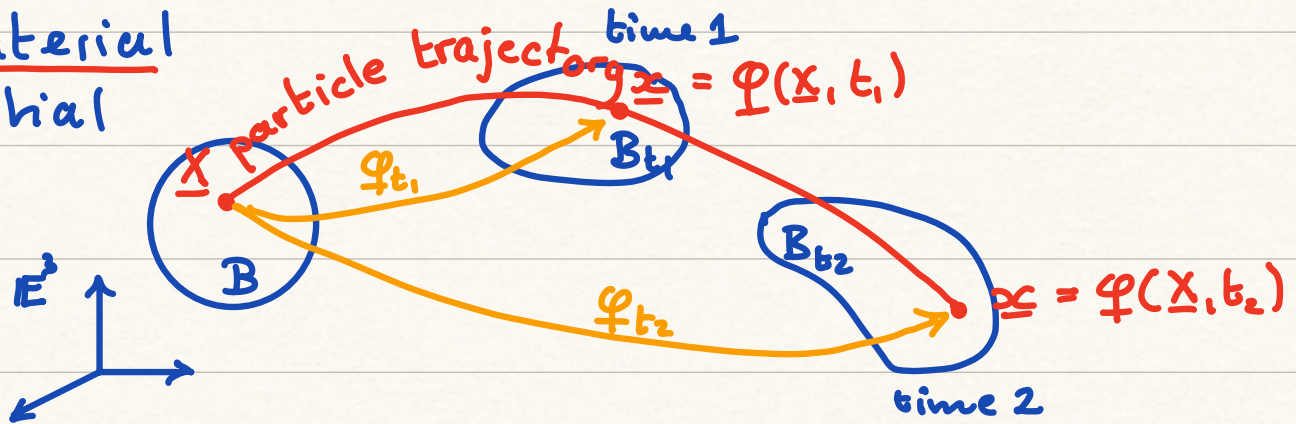


Motion

B_t : • deformed
• spatial
• current

B : • undeformed
• reference
• material
• initial



Continuous deformation of a body over time

$\Rightarrow$ motion: $x = \varphi(\underline{X}, t)$

Inverse motion: $\underline{X} = \psi(x, t)$

Two types of time derivatives:

1) spatial/local: $\frac{\partial \phi}{\partial t} \Big|_x \equiv \frac{\partial \phi}{\partial t}$ (keep x fixed)

rate of change of ϕ as seen by an observer at x .

2) material/total: $\frac{\partial \phi}{\partial t} \Big|_{\underline{X}} \equiv \dot{\phi}$ (keep $\underline{X}$ fixed)

rate of change of ϕ seen by an observer following the path line of a particle.

Velocity and acceleration fields

The velocity and acceleration of a material particle labeled $\underline{x}$ in $\mathcal{B}$ at time t due to motion $\varphi(\underline{x}, t)$

$$\begin{aligned}\underline{V}(\underline{x}, t) &= \dot{\varphi}(\underline{x}, t) = \frac{\partial}{\partial t} \varphi(\underline{x}, t) = \frac{\partial \underline{x}}{\partial t} \Big|_{\underline{x}} \\ \underline{A}(\underline{x}, t) &= \dot{\underline{V}}(\underline{x}, t) = \ddot{\varphi}(\underline{x}, t) = \frac{\partial^2}{\partial t^2} \varphi(\underline{x}, t) = \frac{\partial^2 \underline{x}}{\partial t^2} \Big|_{\underline{x}}\end{aligned}$$

Easy because $\varphi = \varphi(\underline{x}, t) \rightarrow$ material field

Spatial descriptions: $\underline{x} = \underline{\psi}(\underline{x}, t)$

$$\underline{v}(\underline{x}, t) = \underline{V}(\underline{\psi}(\underline{x}, t), t) = \dot{\varphi}(\underline{x}, t) \Big|_{\underline{x} = \underline{\psi}(\underline{x}, t)}$$

$$\underline{a}(\underline{x}, t) = \underline{A}(\underline{\psi}(\underline{x}, t), t) = \ddot{\varphi}(\underline{x}, t) \Big|_{\underline{x} = \underline{\psi}(\underline{x}, t)}$$

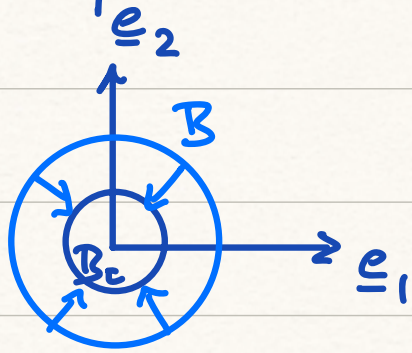
The spatial fields $\underline{v}$ and $\underline{a}$ correspond to the material particle whose current coordinates are $\underline{x}$ at time t .

Note:

material field: $\Pi(\underline{x}, t) \rightarrow \Pi(\underline{\psi}(\underline{x}, t), t) \equiv \Pi_s(\underline{x}, t)$
spatial representation

spatial field: $s(\underline{x}, t) \rightarrow s(\varphi(\underline{x}, t), t) \equiv s_m(\underline{x}, t)$
material representation

Example: Shrinking sphere



$$\varphi(\underline{x}, t) = e^{-\lambda t} \underline{x} \quad \lambda > 0$$

$$\psi(\underline{x}, t) = e^{\lambda t} \underline{x}$$

Velocity & acceleration:

$$\left. \begin{aligned} \underline{V}(\underline{x}, t) &= \dot{\varphi}(\underline{x}, t) = -\lambda e^{-\lambda t} \underline{x} \\ \underline{A}(\underline{x}, t) &= \dot{\underline{V}}(\underline{x}, t) = \lambda^2 e^{-\lambda t} \underline{x} \end{aligned} \right\} \text{material fields}$$

spatial representation $\Rightarrow \underline{x} = \psi(\underline{x}, t) = e^{\lambda t} \underline{x}$

$$\underline{v}(\underline{x}, t) = \underline{V}(\psi(\underline{x}, t), t) = -\lambda e^{-\lambda t} (e^{\lambda t} \underline{x}) = -\lambda \underline{x}$$

$$\underline{a}(\underline{x}, t) = \underline{A}(\psi(\underline{x}, t), t) = \lambda^2 e^{-\lambda t} (e^{\lambda t} \underline{x}) = \lambda^2 \underline{x}$$

Note: $\frac{\partial \underline{v}}{\partial t} = \underline{0} \neq \underline{a} \Rightarrow \underline{a} \neq \frac{\partial \underline{v}}{\partial t}$

spatial acceleration is not local derivative of spatial velocity field.



each car is decelerating as it approaches ACL: $\underline{a}(\underline{x}, t) < 0$

all cars go ≈ 50 mph when they pass green person: $\dot{\underline{v}}(\underline{x}, t) \approx 0$

Compute material derivative of spatial field?

Example: $T(\underline{x}, t) = \alpha t |\underline{x}| e^{\lambda t}$

$$\dot{T}(\underline{x}, t) = \frac{\partial}{\partial t} T(\underbrace{\varphi(\underline{x}, t), t}_{\text{II}}) \Big|_{\underline{x} = \underline{\varphi}(\underline{x}, t)} = \frac{\partial}{\partial t} T_m(\underbrace{\underline{X}, t}_{\text{I}}) \Big|_{\underline{x} = \underline{\varphi}(\underline{x}, t)}$$

Note: spatial representation of material derivative of a spatial field.

I) Differentiate material representation

- pull it back to material configuration

$$\text{motion: } \underline{x} = \varphi(\underline{X}, t) = e^{-\lambda t} \underline{X}$$

$$T(\varphi(\underline{X}, t), t) = \alpha t |e^{-\lambda t} \underline{X}| e^{\lambda t} = \alpha t |\underline{X}|$$

$$T_m(\underline{X}, t) = \alpha t |\underline{X}| \quad \text{material representation}$$

- material time derivative:

$$\dot{T}_m = \frac{\partial}{\partial t} T_m(\underline{X}, t) = \alpha |\underline{X}|$$

- push forward to spatial configuration

$$\text{inverse motion: } \underline{X} = \psi(\underline{x}, t) = e^{\lambda t} \underline{x}$$

$$\dot{T}_m \Big|_{\underline{X} = \psi(\underline{x}, t)} = \alpha |e^{\lambda t} \underline{x}| = \alpha e^{\lambda t} |\underline{x}| = \dot{T}(\underline{x}, t)$$

$$\underline{\dot{T}(\underline{x}, t) = \alpha e^{\lambda t} |\underline{x}|}$$

What if we don't have φ and Ψ ?

II, Use chain rule

$\Rightarrow$ 2 time dependencies $T(\varphi(\underline{x}, t), t)$ & keep $\underline{x}$ fixed!

Quick version:

$$\dot{T}(\underline{x}(t), t) = \frac{\partial T}{\partial t} + \frac{\partial T}{\partial x_i} \underbrace{\frac{\partial x_i}{\partial t}}_{v_i} = \frac{\partial T}{\partial t} + v_i \frac{\partial T}{\partial x_i} = \frac{\partial T}{\partial t} + \underline{v} \cdot \underline{\nabla} T$$

Careful version (keeping track of $\underline{x}$)

$$\dot{T}(\underline{x}, t) = \frac{\partial}{\partial t} T(\underbrace{\varphi(\underline{x}, t)}_{\underline{x}}, t) \Big|_{\underline{x} = \varphi(\underline{x}, t)}$$

$$= \underbrace{\left[\frac{\partial T}{\partial t} \Big|_{\underline{x} = \varphi(\underline{x}, t)} + \frac{\partial T}{\partial x_i} \Big|_{\underline{x} = \varphi(\underline{x}, t)} \frac{\partial \varphi_i(\underline{x}, t)}{\partial t} \right]}_{\dot{T}_m(\underline{x}, t) \text{ material representation}} \Big|_{\underline{x} = \varphi(\underline{x}, t)}$$

velocity: $\frac{\partial \varphi_i}{\partial t} = v_i(\underline{x}, t) = v_i(\underline{x}, t) \Big|_{\underline{x} = \varphi(\underline{x}, t)}$

$$\dot{T}_m(\underline{x}, t) = \frac{\partial T}{\partial t}(\underline{x}, t) + \underbrace{\frac{\partial T}{\partial x_i} v_i(\underline{x}, t)}_{\underline{v}(\underline{x}, t) \cdot \underline{\nabla} T(\underline{x}, t)} \Big|_{\underline{x} = \varphi(\underline{x}, t)}$$

$$\dot{T}(\underline{x}, t) = \underline{v} \cdot \underline{\nabla} T + \frac{\partial T}{\partial t}$$

Example: $T(\underline{x}, t) = \alpha t e^{\lambda t} |\underline{x}|$

$$\underline{v}(\underline{x}, t) = -\lambda \underline{x}$$

$$\frac{\partial T}{\partial t} = \alpha |\underline{x}| (e^{\lambda t} + \lambda t e^{\lambda t}) = \alpha |\underline{x}| (1 + \lambda t) e^{\lambda t}$$

$$\nabla T = T_{,i} \underline{e}_i = \alpha t e^{\lambda t} (x_k x_k)^{1/2},_{,i} \underline{e}_i$$

$$= \alpha t e^{\lambda t} \frac{1}{2} (x_k x_k)^{-1/2} 2(x_{k,i} x_k) \underline{e}_i$$

$$= \alpha t e^{\lambda t} \frac{1}{|\underline{x}|} \delta_{ki} x_k \underline{e}_i = \alpha t e^{\lambda t} \frac{\underline{x}}{|\underline{x}|}$$

$$\dot{T} = \frac{\partial T}{\partial t} + \underline{v} \cdot \nabla T$$

$$= \alpha |\underline{x}| (1 + \lambda t) e^{\lambda t} - \lambda \underline{x} \cdot \left(\alpha t e^{\lambda t} \frac{\underline{x}}{|\underline{x}|} \right) \\ - \alpha \lambda t e^{\lambda t} \frac{\underline{x} \cdot \underline{x}}{|\underline{x}|}$$

$$= \alpha |\underline{x}| e^{\lambda t} (1 + \lambda t - \lambda t)$$

$$\dot{T} = \alpha |\underline{x}| e^{\lambda t} \quad \text{same as method I } \nabla$$

Spatial rep. of mat. deriv. of spatial field:

$$\dot{\phi} = \frac{\partial \phi}{\partial t} + \nabla \phi \cdot \underline{v}$$

and

$$\dot{\underline{\omega}} = \frac{\partial \underline{\omega}}{\partial t} + (\nabla \underline{\omega}) \underline{v}$$

The result for $\dot{\underline{\omega}}$ follows by applying the scalar result to ω_i .

This result is important because it allows the computation of $\dot{\phi}$ and $\dot{\omega}$ without knowledge of ϕ , if the velocity is known!

$\Rightarrow$ in fluid mechanics you never see ϕ

In fluid mechanics: $(\nabla \underline{\omega}) \underline{v} = \underline{v} \cdot \nabla \underline{\omega}$ see HW7!

The spatial acceleration field is defined as

$$\underline{a} = \dot{\underline{v}} = \frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \underline{v}$$

Spatial acceleration is a non-linear function of spatial velocity $\Rightarrow$ basic non-linearity in fluid mechanics

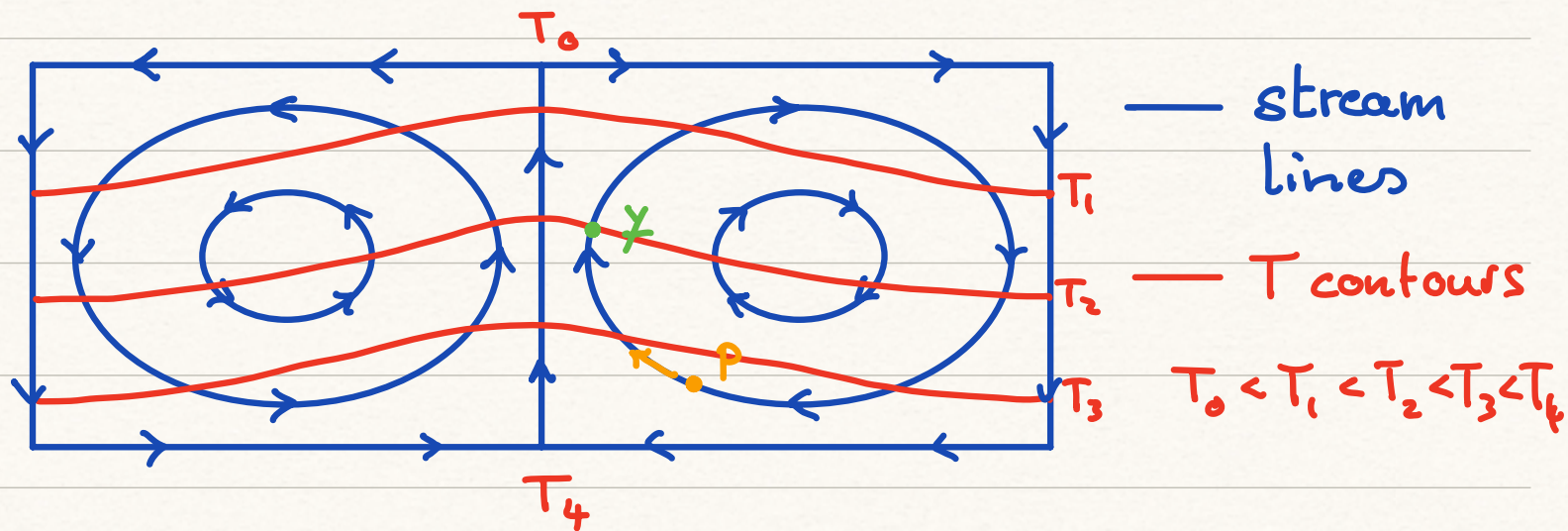
Example: $\underline{v} = -\lambda \underline{x}$ $\underline{a} = \lambda^2 \underline{x}$

$$\underline{a} = \frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \underline{v} \quad \nabla \underline{v} = v_{i,j} \underline{e}_i \otimes \underline{e}_j$$

$$\underline{a} = -\lambda \underline{I} (-\lambda \underline{x}) \quad v_{i,j} = -\lambda x_{i,j} = -\lambda \delta_{ij}$$

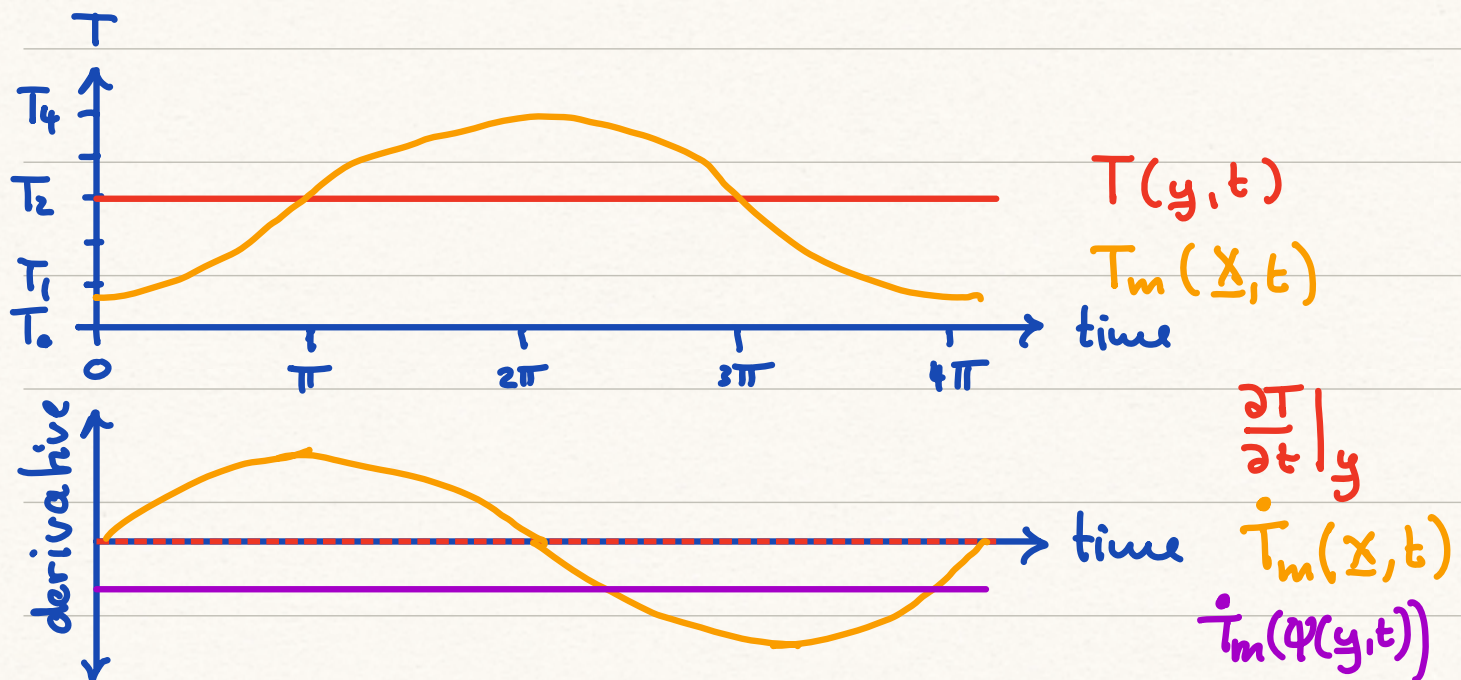
$$\underline{a} = \lambda^2 \underline{x} \quad \Rightarrow \nabla \underline{v} = -\lambda \underline{I}$$

Example: Steady convection



Steady state: $T(\underline{x}, t) = T(\underline{x})$ (spatial field)

Clearly, $T(\underline{y}) = T_2$ and $\frac{\partial T}{\partial t} \big|_{\underline{y}} = 0$



Consider a particle P with initial location $\underline{x}$ and current position $\underline{x} = \varphi(\underline{x}, t)$, which passes

through y once every overturn.

Its temperature is $T(\varphi(\underline{x}, t)) = T_m(\underline{x}, t)$ is a material field and oscillates periodically. Its derivative the "material derivative" is $\dot{T}_m(\underline{x}, t) \neq 0$